



A sorghum genome-wide association study (GWAS) identifies a WRKY transcription factor as a candidate gene underlying sugarcane aphid (*Melanaphis sacchari*) resistance

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Received: 6 October 2021 / Accepted: 19 December 2021 / Published online: 12 January 2022
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Abstract

Main conclusion A WRKY transcription factor identified through forward genetics is associated with sorghum resistance to the sugarcane aphid and through heterologous expression reduces aphid populations in multiple plant species.

Abstract Crop plant resistance to insect pests is based on genetically encoded traits which often display variability across diverse germplasm. In a comparatively recent event, a predominant sugarcane aphid (SCA: *Melanaphis sacchari*) biotype has become a significant agronomic pest of grain sorghum (*Sorghum bicolor*). To uncover candidate genes underlying SCA resistance, we used a forward genetics approach combining the genetic diversity present in the Sorghum Association Panel (SAP) and the Bioenergy Association Panel (BAP) for a genome-wide association study, employing an established SCA damage rating. One major association was found on Chromosome 9 within the *WRKY transcription factor 86* (*SbWRKY86*). Transcripts encoding *SbWRKY86* were previously identified as upregulated in SCA-resistant germplasm and the syntenic ortholog in maize accumulates following *Rhopalosiphum maidis* infestation. Analyses of *SbWRKY86* transcripts displayed patterns of increased SCA-elicited accumulation in additional SCA-resistant sorghum lines. Heterologous expression of *SbWRKY86* in both tobacco (*Nicotiana benthamiana*) and Arabidopsis resulted in reduced population growth of green peach aphid (*Myzus persicae*). Comparative RNA-Seq analyses of Arabidopsis lines expressing *35S:SbWRKY86-YFP* identified changes in expression for a small network of genes associated with carbon–nitrogen metabolism and callose deposition, both contributing factors to defense against aphids. As a test of altered plant responses, *35S:SbWRKY86-YFP* Arabidopsis lines were activated using the flagellin epitope elicitor, flg22, and displayed significant increases in callose deposition. Our findings indicate that both heterologous and increased native expression of the transcription factor *SbWRKY86* contributes to reduced aphid levels in diverse plant models.

Keywords Plant defense · Herbivory · Aphid · WRKY transcription factor · Callose

Communicated by Anastasios Melis.

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Introduction

All plants in ecological and agricultural settings face a diverse array of stresses imposed by biological agents. Combined pressures of native and invasive insect pests commonly reduce crop yields by 15% annually (Bradshaw et al. 2016). The sugarcane aphid (SCA: *Melanaphis sacchari*) has long been known to occur on sorghum (*Sorghum bicolor*) as a minor pest; however, in 2013, a specific predominant SCA haplotype became an aggressive pest on grain sorghum in the United States (Armstrong et al. 2015; Harris-Shultz et al. 2017; Sharma et al. 2013). As a closely related genetic model to maize, sorghum is of shared agronomic significance for food, animal feed, pasture fodder, biofuel feedstock and is increasingly important for understanding the basis of drought-tolerance mechanisms (Mace et al. 2013). As a threat to crop production, SCA now infests all sorghum-growing areas in the U.S., causing yield losses between 10 and 50% in fields lacking insecticide treatments (Bowling et al. 2016; Szczepaniec 2018). High populations of SCA in sorghum result in both direct injury to plants and indirect injury stemming from honeydew and subsequent epiphytic sooty mold growing on excreted carbohydrates that collectively reduce photosynthetic capacity.

Plants recognize herbivores through both damage and herbivore-associated molecular patterns, initiating intricate signaling that triggers a complex array of constitutive and inducible defenses for protection (Howe and Jander 2008). Within the scope of insect herbivory, piercing-sucking insects such as aphids utilize stylets to extract phloem sap, causing extremely low levels of direct plant tissue damage and associated defense signals (Nalam et al. 2019). Despite the modest wounding caused by aphid attack, activation of oxylipin phytohormone signaling including 12-oxo-11-phytodienoic acid (12-OPDA) and JA are broadly implicated in mounting successful defenses against aphids (Grover et al. 2020; Gao et al. 2007). Many plant genes impacting aphid resistance have been discovered through targeted mutant analyses and over-expression studies (Singh et al. 2011; Rossi et al. 1998; Mondal et al. 2018; Meihls et al. 2013; Louis et al. 2010; Chung et al. 2021). In *Arabidopsis*, the gene *PHYTOALEXIN DEFICIENT4* (*PAD4*), encodes a lipase which regulates *Myzus persicae* feeding; *pad4* knockout mutants are more susceptible to aphid attack, whereas plants over-expressing *PAD4* display enhanced aphid resistance (Pegadaraju et al. 2007; Louis and Shah 2015; Dongus et al. 2020; Lei et al. 2014). Similarly, the aphid-regulated *Arabidopsis* gene *Myzus persicae*-induced lipase 1 (*MPL1*) displayed a similar pattern of increased *M. persicae* populations on *mpl1* mutants and reduced populations on *35S::MPL1*

over-expression lines (Louis et al. 2010). Heterologous over-expression of the mannose-binding *Allium sativum* leaf agglutinin (*ASAL*) in rice and chickpea imparts enhanced defense against multiple species of sap-sucking insects (Chakraborti et al. 2009; Saha et al. 2006). The tomato nucleotide-binding domain, leucine-rich repeat (NLR) *Mi* gene can confer resistance to specific biotypes of the potato aphid (*Macrosiphum euphorbiae*) and similarly the NLR *Vat* gene in muskmelon (*Cucumis melo*) mediates resistance to *Aphis gossypii* (Rossi et al. 1998; Boissot et al. 2016). Beyond single defense-related genes, a number of WRKY family transcription factors (TFs) in diverse plant species have been demonstrated to fine-tune complex defense outputs and modulate aphid population growth through both gene-silencing and over-expression studies (Atamian et al. 2012; Li et al. 2015, 2016). The presence and concentration of plant-specialized metabolites can also influence aphid populations. In maize, resistance to the corn leaf aphid (*Rhopalosiphum maidis*) is genetically associated with a cluster of specific O-methyl transferase genes that impact levels of the benzoxazinoid (BX) 2,4-dihydroxy-7-methoxy-1,4-benzoxazin-3-one (DIMBOA) (Meihls et al. 2013). Generally considered as a direct defense, reduced DIMBOA levels in BX-deficient *bx1* mutants increase susceptibility to the bird cherry-oat aphid (*Rhopalosiphum padi*); however, DIMBOA further functions as a signal to enhance chitosan-induced callose deposition (Ahmad et al. 2011). Diverse plant-specialized metabolites can regulate aphid resistance with examples spanning BX, cardenolides, alkaloids, and glucosinolate derivatives (Kim and Jander 2007; Züst and Agrawal 2016). Independent of the various host plant resistance or tolerance mechanisms (Mitchell et al. 2016) within these examples, the identification of natural genetic variation in insect resistance is desirable to speed breeding focused on the application of non-transgenic pest control approaches.

To date, multiple approaches have been used to assess sorghum defense and tolerance mechanisms with the intended goal of understanding improved aphid resistance in the field (Grover et al. 2020; Wang et al. 2013; Zhu-Salzman et al. 2004). Using a biparental mapping population of recombinant inbred lines (RILs) derived from SCA-susceptible and -resistant parents, BTx623 and Henong16, respectively, SCA resistance was localized to a narrow locus on chromosome 6, termed *Resistance to Melanaphis sacchari* 1 (*RMES1*) (Wang et al. 2013). The *RMES1* locus contains five genes, namely *Sb06g001620*, *Sb06g001630*, *Sb06g001640*, *Sb06g001645* and *Sb06g001650*, hypothesized to encode an innate immunity-associated WD40 protein, an RNA-binding protein, and 3 different NBS-LRR-resistance proteins (Wang et al. 2013). Further studies leveraging a sorghum RIL population derived from the SCA-resistant line 407B and SCA-susceptible line 7B were used to confirm the *RMES1* locus

and identify 3 additional associated loci on chromosome 6 (Zhang et al. 2021). Copy number variation and expression variation in the *RMES1* locus NBS-LRR genes were highlighted for predicted causal effects on SCA resistance loci (Zhang et al. 2021). Using a RIL population between Tx2783 and BTx623, further analyses detected sequence variation in disease resistance (R) genes that segregate with an SCA tolerance quantitative trait locus (QTL) (Wang et al. 2021). Useful sources of SCA resistance have been observed in many different sorghum lines; however, the precise genes responsible commonly remain unknown (Paudyal et al. 2019; Limaje et al. 2018; Hayes et al. 2019; Armstrong et al. 2015).

Collectively, hundreds of studies in sorghum have been conducted leveraging Quantitative Trait Loci (QTL) mapping, and increasingly genome-wide association studies (GWAS) are used to connect genes to diverse phenotypes of interest. While traits such as seed size, flowering time, height, and others have been emphasized, over 220 different sorghum traits have been examined, resulting in over 6000 QTLs and an array of underlying candidate genes (Mace et al. 2019). Sorghum GWAS have been used to understand gene candidates regulating development time, seed protein, fat, starch, polyphenol levels and other traits relevant to production (Boyles et al. 2016; Brenton et al. 2016; Fernandez et al. 2015; Morris et al. 2013; Rhodes et al. 2014; Sukumaran et al. 2012). Increasingly expansive sorghum resources now exist and include over 10,000 germplasm lines that have been genotyped for 459,304 single-nucleotide polymorphism (SNP) markers (Hu et al. 2019).

In an effort to identify gene candidates underlying SCA resistance, we conducted a large field-based GWAS using foliar SCA damage as a phenotypic trait in two large global sorghum diversity panels, namely the Sorghum Association Panel (SAP) (Morris et al. 2013; Casa et al. 2008) and the Bioenergy Association Panel (BAP) (Brenton et al. 2016), for which extensive genotype data exist. GWAS analyses identified a locus on chromosome 9 as significantly associated with SCA resistance. From this locus, the transcription factor *SbWRKY86* (Sobic.009G238200) emerged as a promising gene candidate with a role in aphid elicited responses supported by multiple sorghum transcriptomic studies and demonstrated by increased expression in SCA-resistant varieties as compared to susceptible lines. Heterologous expression of *SbWRKY86* protein in two plant models conferred reduced population increases in the aphid *Myzus persicae*. Analyses of *SbWRKY86*-regulated changes in gene expression implicated callose production as a potential underlying factor for the observed enhanced resistance, and plants expressing *SbWRKY86* protein displayed increased callose deposition upon immune activation. Together our data support *SbWRKY86* as a gene candidate underlying variation in sorghum resistance to SCA and highlight consideration of

germplasm with elevated *SbWRKY86* expression levels for use in breeding for enhanced aphid resistance.

Materials and methods

Plant and insect materials and growth conditions

Sorghum lines were provided by the USDA Germplasm Resources Information Network (GRIN). For controlled SCA assays, sorghum seeds were germinated in BM2 potting soil (Berger; Saint-Modeste, QC, Canada) supplemented with 18-18-21 Tomato Plant Food fertilizer (Miracle-Gro, <https://www.miraclegro.com/>) and grown in a greenhouse under conditions described previously (Ding et al. 2020). *Arabidopsis thaliana* seeds were surface-sterilized using chlorine (Cl₂) gas, stratified for 48 h at 4 °C and then germinated in plates containing half-strength MS media (Murashige and Skoog 1962) (PhytoTechnology Laboratories), adjusted to pH 5.8 using KOH with 0.8% phytigel (PhytoTechnology Laboratories). *Arabidopsis* seedlings were either kept in plates or transferred to BM2 soil. *Nicotiana benthamiana* seeds were germinated directly in BM2 soil supplemented with 20-20-20 General Purpose fertilizer (Jack's Professional, <https://www.jrpeters.com/>). Both *Arabidopsis* and *N. benthamiana* plants were cultivated in a growth room at 22 ± 2 °C, with 150 µmol m⁻² s⁻¹ light for either 16 (*Arabidopsis*) or 12 hours (*N. benthamiana*). A colony of *Melanaphis sacchari* originated from Kettleman City, California was obtained from Dr. Spencer Walse (USDA-ARS, Parlier CA) and reared on 3- to 6-week-old sorghum plants. The tobacco-adapted red lineage of *Myzus persicae* (Ramsey et al. 2014) was obtained from Dr. Georg Jander (Boyce Thompson Research Institute) and reared on 4- to 6-week-old *A. thaliana* ecotype Columbia-0 (Col-0) plants.

Association analyses

Seeds from the Sorghum Association Panel (SAP) (Casa et al. 2008) and sorghum Bioenergy Association Panel (BAP) (Brenton et al. 2016) obtained from GRIN were planted at Gibbs Research Farm near Tifton, GA (USA) (31.5 °N, 83.5 °W) on June 9, 2017. A randomized field design was employed using short rows of 20 plants per inbred line with natural aphid infestation of the test plots. SCA damage ratings were scored using 9-week-old plants on August 13–14, 2017. Plants were rated for severity of damage using a 1–9 scale developed by Sharma et al. (2013) and broadly used by others (Armstrong et al. 2015). Three plants from each line were used in the consideration and scoring of SCA damage estimates using the top fully expanded leaves and bottom green leaves. SCA ratings

spanned a ranking of (1) few aphids present on the lower one to two leaves, with no apparent damage to the leaves to (9) heavy aphid infestation up to the flag leaf, and 80% of the leaves showing aphid damage (drying-up symptoms), and heavy honeydew/sooty mold on the leaves and on the soil below (Sharma et al. 2013). The combined SNP marker map for the sorghum BAP and SAP were filtered to a total of 76,609 marker SNPs through the removal of sites with minimum 20% frequency and maximum 80% frequency using TASSEL5 (v5.2.74) (Bradbury et al. 2007). The filtered SNP marker map was then imputed using LinkImpute (v1.1.5, default settings) (Money et al. 2015). Prior to the association analysis, the SCA damage rating score was normalized using the Ordered Quantile normalizing transformation method that was selected by bestNormalize (v1.8.2). Association analysis on the imputed SNP marker map was conducted by Memory-efficient, Visualization-enhanced, and Parallel-accelerated R package (rMVP; v1.0.6) using three algorithms: General Linear Model (GLM; PCA), Mixed Linear Model (MLM; PCA + K) and Fixed and random model Circulating Probability Unification (FarmCPU) (Yin et al. 2021; Liu et al. 2016). To account for population structure, the first three principal components and the kinship matrix were used. The Efficient Mixed-Model Association (EMMA) method was used for the MLM variance component analysis.

Comparative genomic analyses

A genomic comparison between the sorghum SCA resistance locus and the corresponding maize syntenic region was conducted through two approaches: CoGe and genoPlotR (v0.8.11) (Lyons and Freeling 2008; Guy et al. 2010). To generate the genoPlotR synteny plot, the NCBI standalone blastn software (word_size 20, e-value 1) was used to compare the genomic sequences of sorghum and maize between the left border (LB—*Sobic.009G238100*) and right border (RB—*Sobic.009G238500*) genes (Camacho et al. 2009). To generate a phylogeny of all sorghum, maize and Arabidopsis WRKY genes, the predicted protein sequences from primary transcripts (from Phytozome) were annotated and all WRKY domain-containing genes identified by PlantTFDB (Goodstein et al. 2012; Jin et al. 2017). The annotated WRKY protein sequences were aligned with famsa (v1.6.2, default parameters), clipped using ClipKIT (v1.1.5, default parameters) and a phylogenetic tree was constructed using IQ-TREE (v2.1.4, default parameters and 1,000 bootstrap replications) (Deorowicz et al. 2016; Nguyen et al. 2015; Steenwyk et al. 2020). The phylogenetic tree was visualized with FigTree (<http://tree.bio.ed.ac.uk/software/figtree/>).

qRT-PCR analysis of *SbWRKY86* expression

Total RNA was isolated from sorghum leaves with TRIzol reagent, treated with DNase according to the manufacturer's instructions (Life Technologies), and cDNA synthesized using MMLV Reverse Transcriptase and random decamer primers (Thermo Fisher Scientific). Gene expression analysis by qRT-PCR was performed using cDNA generated from 33.3 ng RNA as template, SsoAdvanced (tm) Universal SYBR(R) Green Supermix (Bio-Rad) and a StepOne Real-Time PCR System (Applied Biosystems). Based on the threshold cycle (Ct) value, ΔC_t was calculated relative to *Glyceraldehyde 3-Phosphate Dehydrogenase* (*GAPDH*, *Sobic.009G016700*) as a reference gene. Transcript abundance was calculated relative to its corresponding untreated control. All primers used are shown in Supplementary Table 1.

Generation of plasmids and plants for heterologous expression of *SbWRKY86*

To generate the desired constructs, the coding region of *SbWRKY86* (*Sobic.009G238200*) from *Sorghum bicolor* line BTx623 was amplified from cDNA via standard PCR and cloned into the pENTR/D-TOPO vector (Invitrogen). Primers used are shown in Supplementary Table 1. Expression vectors driven by the CaMV 35S promoter and producing C-terminal yellow fluorescent protein (YFP) fusions were generated through Gateway cloning in the pGWB441 destination vector (Nakagawa et al. 2007). Constructs were verified by Sanger DNA sequencing and transformed into *Agrobacterium tumefaciens* strain GV3101 (pMP90). Arabidopsis Col-0 plants were transformed with *A. tumefaciens* carrying the 35S::*SbWRKY86-YFP* expression construct by the floral-dip method (Clough and Bent 1998), and successful transformants were selected on 1/2 × MS media supplemented with 100 $\mu\text{g mL}^{-1}$ kanamycin. All experiments used plants from homozygous T₂ and T₃ generations that were confirmed to express *SbWRKY86-YFP* protein through western blotting. For transient expression of *SbWRKY86-YFP* in *Nicotiana benthamiana*, leaves were infiltrated with *A. tumefaciens* carrying the vectors as described (Sparkes et al. 2006). Experiments were performed 24 h post-agroinfiltration and *SbWRKY86-YFP* protein expression was confirmed through western blotting.

Western blotting to assess heterologous *SbWRKY86-YFP* protein expression

Leaves from 4-week-old Arabidopsis plants heterologously expressing 35S::*SbWRKY86-YFP* or leaf discs from *N. benthamiana* plants agroinfiltrated with 35S::*SbWRKY86-YFP* constructs were harvested and frozen in liquid nitrogen.

Tissue was ground with a prechilled mortar and pestle, homogenized in Laemmli buffer (0.24 M Tris-HCl pH 6.8, 6% SDS, 30% glycerol, 16% β -mercaptoethanol, 0.006% bromophenol blue, 10 M urea) using a 1:1.5 (mass/volume) tissue to buffer ratio and boiled at 95 °C for 5 min. The samples were then centrifuged at 13,000 $\times$ g for 5 min at room temperature, and the supernatant collected for western blotting. Proteins were separated by SDS-PAGE. Anti-GFP (1:1000, Thermo Scientific) antibody and anti-rabbit-HRP-conjugated secondary antibody (1:2000, Sigma) were used with the Super Signal West Pico Maximum Chemiluminescent detection kit (Thermo Scientific) to detect protein expression.

Plant-aphid bioassays

All bioassays were performed as no-choice tests, with specific details for each species studied as follows: Germplasm for sorghum inbred comparisons were selected based on SCA-susceptible (PI534135, PI597950) and -resistant (PI656011, PI656043) phenotypes observed in the 2017 Tifton, GA field trials. Two-week-old greenhouse-grown sorghum plants were infested with a single wingless *M. sacchari* adult and caged in a mesh enclosure to prevent insect movement among plants. Aphids were counted daily up to 5 days, with 8–10 replicate plants used for each sorghum genotype. For Arabidopsis, 4-week-old Col-0 or *CaMV* 35S::*SbWRKY86-YFP*-expressing plants were infested with a single wingless *M. persicae* adult and caged in a mesh enclosure. After 4 days, aphids were counted on each of the 16–18 replicate plants used per genotype. For *N. benthamiana*, leaves from 3- to 4-week-old *N. benthamiana* plants were infiltrated with *A. tumefaciens* carrying the *CaMV* 35S::*SbWRKY86-YFP* construct as described previously (Sparkes et al. 2006). 24 h post-agroinfiltration, cages were placed on the abaxial side of infiltrated leaves, and four tobacco-adapted wingless *M. persicae* adults (Ramsey et al. 2014) were placed in each cage. Plants agroinfiltrated with an empty vector were used as controls. Aphids were counted after 7 days with 15 replicate leaves used per construct.

Preparation of RNA samples and generation of RNA-Seq data

To generate sufficient tissue and samples for RNA-Seq analyses of sorghum BTx623 responses to SCA attack, the most apical expanded leaves of 40-day-old plants were infested with 10 adult SCA confined to a localized mesh cage following the methodology of Tzin et al. (2015). Upon initiation of the experiment, mesh cages were affixed to leaves of all plants simultaneously, and the introduction of SCA to the cages was staggered such that all tissues for gene expression analysis were harvested on the same day and within a

1-h interval (3 to 4 PM), 48 h after the start of the experiment. The design resulted in a 0-, 2-, 4-, 8-, 24-, and 48-h time course with reduced circadian influence. Eight plants were used for each treatment and time point; however, for gene expression analysis two individual plants/leaf sections were pooled to create each final biological replicate ($n=4$ per time point). RNA-Seq analysis was performed by the Department of Energy Joint Genome Institute (DOE-JGI) using Illumina HiSeq 2500 to generate single-end reads. The raw RNA-seq reads were deposited to the NCBI Sequence Read Archive (SRA) and are available under the BioProject accession number PRJNA786558. For study of altered basal gene expression in Arabidopsis plants expressing *p35S::WRKY86-YFP*, plants were grown on soil for 25 days. Two independent insertional lines, 13 and 14, were used in comparison to Col-0 wild type as a control. Each biological replicate consisted of individual leaves from four separate plants pooled together. Leaf tissues were harvested as described (Tzin et al. 2015). Wild-type Col-0 plants were used as controls. Total RNA was isolated with the Spectrum Plant Total RNA kit (Sigma) and subsequently DNase-treated using the Turbo DNA-free kit (Ambion). Preparation of RNA-Seq data was performed by Novogene Corporation. In brief, the RNA quality and quantity was analyzed using the Agilent Bioanalyzer 2100 (Agilent technologies, Santa Clara, CA, US), a cDNA library was constructed and sequenced using Illumina HiSeq platform PE150, to generate paired-end reads that were used for analysis. The raw RNA-seq reads were deposited to the NCBI Sequence Read Archive (SRA) and are available under the BioProject accession number PRJNA786556.

Analysis of sorghum, maize and Arabidopsis RNA-Seq data

To examine aphid-responsive expression changes for sorghum genes contained in our mapping interval, or for corresponding orthologous maize genes, raw reads generated from the time-course analysis of BTx623 sorghum responses to SCA described above and raw reads obtained from the NCBI SRA BioProjects PRJNA451032 (sorghum) and PRJNA295410 (maize) were analyzed (Tzin et al. 2015; Kiani and Szczepaniec 2018). To conduct these analyses, the GENOZIP (v12.0.31) compressed paired-end reads were filtered using FASTP (v0.20.1, default settings), aligned to the sorghum (v3.1.1) or maize (B73 RefGen_v4) reference genomes (from Phytozome) using BWA-MEM (v0.7.17-r1188, default settings) and sorted using SAMBAMBA (v0.8.0, default settings) (Chen et al. 2018; Goodstein et al. 2012; Lan et al. 2021; Li 2013; Tarasov et al. 2015). FeatureCounts was used to generate the raw count data for all samples (v2.0.1, default settings) (Liao et al. 2014). The raw count data for each experiment were analyzed using

edgeR (v3.34.0) to generate CPM data and to calculate the $\log_2(\text{fold-change})$ with adjusted P values using the Exact-Test method (Robinson et al. 2010). The R package ComplexHeatmap (v 2.7.10) was used to visualize the results (Gu et al. 2016). Bioinformatic analysis of Arabidopsis RNA-Seq data was performed by Novogene Corporation. Raw reads were filtered for adaptor contamination, to eliminate reads more than 10% uncertainty and more than 50% low quality (<5) nucleotides, followed by alignment using HISAT2 (v2.0.5, --dta --phred33) (Kim et al. 2015). FeatureCounts (v1.5.0-p3, default settings) was used to generate read counts from aligned files (Liao et al. 2014). Differential gene expression analysis was performed using DESeq2 (v1.20.0) and the P values were adjusted using the Benjamini and Hochberg's method (Love et al. 2014).

Network analysis of genes differentially expressed in 35S::SbWRKY86-YFP Arabidopsis lines

Differentially expressed genes were defined as those demonstrating a twofold change either up or down with an adjusted P value <0.05 . The set of genes conforming to this definition in both 35S::SbWRKY86-YFP lines 13 and 14 was identified and the STRING Protein–Protein Interaction Networks Functional Enrichment Analysis program was used to infer a protein interaction network for the protein products of these genes (Szklarczyk et al. 2018). The network was imported into Cytoscape for further annotation, analysis and visualization (Shannon et al. 2003) with $\log_2(\text{fold-change})$ used as a gradient for the interior color of the node. The R metapack package was used to combine P values for differential expression of each gene in both lines through Fisher's method (Dewey 2020), and the $\log_{10}$ of these combined P values was represented by the color gradient of node borders. Node edge darkness was assigned as a gradient by the String database score, while node size was assigned by degree of connectivity. Nodes with no degree of connectivity within the network were excluded. Potential hub genes with a high degree of connectivity were identified within the network using the cytoHubba (Chin et al. 2014) plugin in Cytoscape.

Promoter scanning and enrichment analysis of WRKY DNA-binding motifs

The annotated DNA-binding motif of AtWRKY30 (AT5G24110), the closest Arabidopsis ortholog to SbWRKY86, was obtained from PlantTFDB (Jin et al. 2017). With promoter regions defined as a 1 kb interval upstream from the transcription start site (TSS), promoters of the genes significantly upregulated in both independent lines of Arabidopsis heterologously expressing 35S::SbWRKY86-YFP were used for motif scanning and enrichment analysis as compared to the promoter sequences

of all Arabidopsis genes that were not significantly upregulated as a control set. MEME Suite (v5.4.1) FIMO was used for promoter scanning to detect AtWRKY30-binding W-box motif matches within the promoters of SbWRKY86-responsive genes (significance threshold $P < 0.001$) (Grant et al. 2011). Statistical significance of AtWRKY30-binding W-box motif enrichment in the promoters of SbWRKY86-responsive genes compared to promoters of all other genes (default parameters) was calculated using the MEME Suite SEA tool (Bailey and Grant 2021).

Callose quantification

Callose staining and quantification was performed as described previously (Varsani et al. 2019). Briefly, for Arabidopsis experiments, 10–12-day-old seedlings of Col-0 wild type and Arabidopsis transgenic lines expressing 35S::SbWRKY86-YFP were treated with water (control) or 1 μM flg22 for 24 h. For sorghum studies, lines PI 597950 and PI 656043 were grown in greenhouse conditions for 16 days, after which 10 sugarcane aphids were caged approximately 1 cm from the leaf tip on the adaxial side of the leaf for 3 days. Following the 3-day infestation period, aphids were counted and the length of leaf within each cage was stained for callose deposition by adapting a previously published protocol (Schenk and Schikora 2015). For both experiments, leaf samples were placed in 98% (v/v) ethanol for 48 h to clear the chlorophyll, and once the leaves became transparent, they were placed in 70% (v/v) ethanol. After washing three times with distilled water, leaves were stained for 3–4 h in 150 mM K_2HPO_4 (pH 9.5) containing 0.01% (v/v) aniline blue (Sigma-Aldrich). Leaves were mounted on slides using 50% (v/v) glycerol and callose deposits visualized by microscopy using an ultraviolet (UV) filter with a Nikon Eclipse microscope. For Arabidopsis, callose deposits were enumerated per mm^2 of leaf tissue on the adaxial side of each leaf using ImageJ (Schindelin et al. 2012). Trainable Weka Segmentation (TWS) plugin for ImageJ was used to analyze leaf images (Arganda-Carreras et al. 2017). Leaf images were scaled to mm per pixel and areas were calculated for each leaf image. TWS was used to calculate the total number of callose particle counts on the leaf itself. Callose deposits per millimeter squared were produced for each leaf as previously described (Mason et al. 2020). For sorghum, six images were taken for each biological replicate and the average number of callose deposits across the six images was calculated per mm^2 of leaf tissue for each biological replicate.

Gene accession numbers

Accession numbers for all genes referred to in this paper are compiled in Supplementary Table 2.

Statistical analyses

For all experiments, the number of biological replicates is stated in both the respective Material and methods sections and in figure legends. For statistical analyses not described explicitly in other sections, all remaining analyses were conducted using GraphPad Prism 9.2 (Graph-Pad Software, Inc., <https://www.graphpad.com/>). Mean values and standard errors of the Mean (SEM) were calculated, and data analyzed using one-way factorial analysis of variance ($P < 0.05$) followed by Tukey's test corrections for multiple comparison between control and treatment groups. Means were separated using Student's *t* test (unpaired, two-tailed) as indicated in the figure legends. P values < 0.05 were considered significant. For western blot and callose staining images, results shown are representative of replicate experiments.

Results

Genome-wide association study (GWAS) using field SCA assessments identifies a locus on chromosome 9 associated with aphid damage levels

To identify genetic loci and gene candidates associated with lower SCA damage levels in diverse sorghum germplasm, we examined 696 lines (Supplementary Table 3) from two large association panels, namely the Sorghum Association Panel (SAP) and the Sorghum Bioenergy Panel (BAP) (Casa et al. 2008; Brenton et al. 2016). Of the 528,052 SNPs genotyped from these lines, 76,609 marker SNPs were obtained through the removal of sites with less than 20% frequency and greater than 80% frequency. As a field-assessed sorghum trait, SCA damage ratings were analyzed by GWAS using GLM, MLM and FarmCPU methods, which all identified the same chromosome 9 SNP (S9_57630053) as most significantly associated with this trait (Fig. 1a–c). SNP S9_57630053 is located in the gene *Sobic.009G238200*, which encodes a WRKY class transcription factor previously designated as *SbWRKY86* (Fig. 1d) (Baillio et al. 2020). The next most significantly associated SNP in two of the three analyses, S9_57629133, is also located in the *SbWRKY86* gene. SNP S9_57630053 variants encoding a T were more commonly associated with low levels of SCA damage than those encoding a G, whereas for SNP S9_57629133, a G was more commonly associated with low levels of SCA damage as opposed to the A variant (Fig. 1e).

Additional SNPs were significantly associated with SCA resistance only in single analyses; the GLM identified a significant ($P = 6.49 \times 10^{-7}$) SNP (S7_9746662) on chromosome 7 located in *Sobic.007G082300* which encodes a BTB/POZ/MATH-domain-containing protein (Fig. 1d, Supplementary Fig. 1). In contrast, the

FarmCPU method identified additional significant SNPs on chromosome 1 (S1_4120345; $P = 2.93 \times 10^{-7}$) located near *Sobic.001G055100* (F-box domain-containing protein), chromosome 4 (S4_629321251; $P = 4.82 \times 10^{-7}$) located near *Sobic.004G287000* (SEC14 cytosolic factor family protein) and chromosome 10 (S10_45992029; $P = 2.48 \times 10^{-7}$) located near *Sobic.010G156100* (GDLS-like Lipase/Acylhydrolase superfamily protein) (Fig. 1). As the gene colocalized with the most statistically significant and consistent association, we prioritized further consideration of *SbWRKY86*. The closest Arabidopsis ortholog to *Sobic.009G238200* is *AtWRKY30* (At5g24110), which is known to be rapidly transcriptionally regulated by both methyl jasmonate and 12-oxo-11-phytodienoic acid (12-OPDA) (Taki et al. 2005). Multiple studies have implicated *AtWRKY30* in promoting protection against diverse oxidative, biotic and abiotic stresses in both native and heterologous systems (El-Esawi et al. 2019; Machens et al. 2014; Scarpeci et al. 2013; Zou et al. 2019). Notably, the chromosome 9 locus is unlinked to the highly studied *RMES1* SCA resistance locus on chromosome 6 (Wang et al. 2013; Zhang et al. 2021; Muleta et al. 2021).

Transcriptomic analysis of aphid herbivory in sorghum

To consider *SbWRKY86* and neighboring genes at the chromosome 9 locus as potentially associated with SCA-elicited sorghum responses, RNA-Seq was used to analyze transcriptome changes in a time-course study of 7-week-old BTx623 plants infested with *M. sacchari* (Supplementary Table 4). Using these data, SCA-induced changes in gene expression within the chromosome 9 mapping interval were assessed. At 48 h post-SCA-infestation, the overall magnitude of transcriptional change for genes in this locus was modest; however, in comparison to neighboring genes, the largest SCA-induced change was observed for *SbWRKY86* (Fig. 2a). Expression of genes in the mapping interval was further examined using a previously published study; *SbWRKY86* was again the most SCA-responsive gene, with significantly increased transcript levels 48 h post-infestation in the SCA-resistant sorghum line DKS 37-07 (Fig. 2a, Supplementary Tables 4 and 5) (Kiani and Szczepaniec 2018). Interestingly, the same study found that while *SbWRKY86* expression was still SCA-responsive in a susceptible line (44-20), the induced change in expression was significantly lower (Fig. 2b). In both our current data and previous studies (Kiani and Szczepaniec 2018), neighboring genes in the mapping interval displayed modest levels of transcriptional change following SCA infestation and largely encoded proteins of unknown function (Supplementary Tables 4 and 5).

To consider candidate genes in a crop closely related to sorghum, we examined the syntenic region of the maize

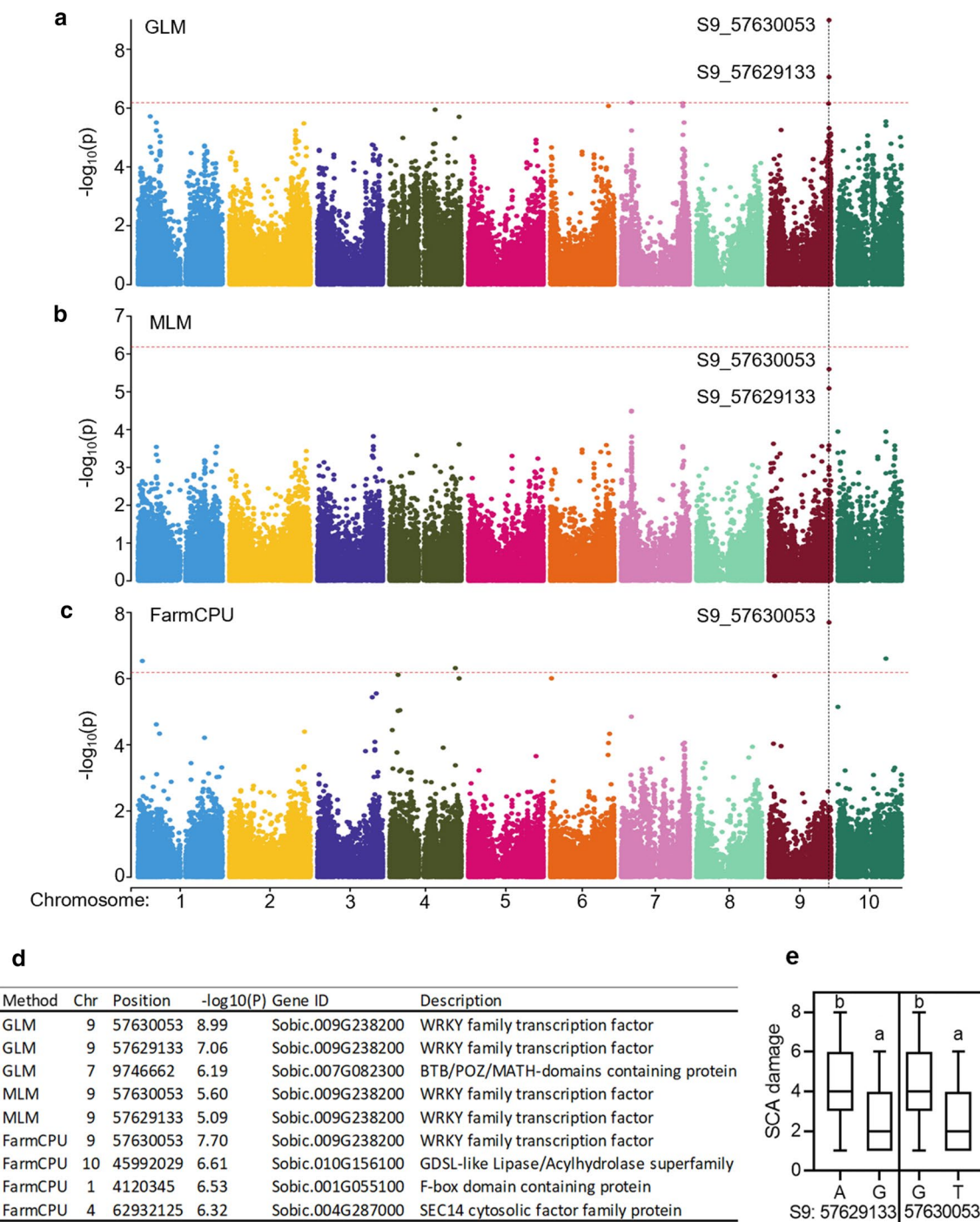


Fig. 1 GWAS to identify loci influencing sugarcane aphid (SCA) damage ratings in sorghum. **a–c** Manhattan plots displaying the significance of associations between single-nucleotide polymorphism (SNPs) and Ordered Quantile normalized transformation of scored SCA damage ratings in the two sorghum association panels, namely the combined SAP and BAP. The *x*-axis represents chromosomal locations and the *y*-axis, $-\log_{10}(P)$ values for genotypic associations obtained using **a** GLM(PCA), **b** MLM(PCA + K) and **c** FarmCPU algorithm models. The horizontal line represents the threshold for Bonferroni adjusted genome-wide significance ($P < 0.05$). The verti-

cal line represents the top significant SNPs associated with SCA on chromosome 9, which were present in Sobic.009G238200, a gene encoding the WRKY transcription factor SbWRKY86. **d** Table of most statistically significant SNPs associated with the SCA damage rating and the nearest corresponding genes detected for each model. **e** Box-plots showing the distribution of the raw sugarcane aphid (SCA) damage score for variants of the two SNPs (S9_57629133, S9_57630053) significantly associated with SCA damage. Different letters (**a**, **b**) represent significant differences ($P < 0.05$) as determined by a pairwise Student's *t* test

genome corresponding to the sorghum SCA resistance-associated locus and identified *ZmWRKY106* (*Zm00001d009619*) as the syntenic maize orthologue of *SbWRKY86* (Fig. 2c, Supplementary Fig. 2, Supplementary Table 6). A protein-derived phylogeny of *SbWRKY86* and related WRKY proteins in sorghum, maize and Arabidopsis confirmed *ZmWRKY106* as the most closely related maize protein and demonstrated that both genes exist within a branch of WRKYs that has proliferated in maize and sorghum and for which the most closely related Arabidopsis WRKY is *AtWRKY30* (Fig. 2d, Supplementary Fig. 3, Supplementary Table 7). The identified maize syntenic region was leveraged to examine gene expression in a maize RNA-Seq dataset generated from leaves infested by the corn leaf aphid, *Rhopalosiphum maidis* (Supplementary Table 8) (Tzin et al. 2015). Within this corresponding region of the maize genome, only expression of *ZmWRKY106* was significantly increased upon attack by *R. maidis* (Fig. 2a). Interestingly, while *ZmWRKY106* expression increased within just a few hours after *R. maidis* infestation, *SbWRKY86* was not observed to significantly increase until 48 h post-infestation with SCA (Fig. 2e). This may be due to variation in regulation between the two crop species, or due to the moderate SCA-susceptibility of sorghum line BTx623 used in our time-course study. Our analyses and consideration of the chromosome 9 SCA resistance-associated locus identified by GWAS (Fig. 1) support *SbWRKY86* (Sobic.009G238200) as a top and re-occurring aphid-responsive candidate gene across diverse datasets (Kiani and Szczepaniec 2018; Tzin et al. 2015) (Fig. 2 and Supplementary Tables 4, 5, 8). As gene candidates, WRKY family of transcription factors are of broad interest given their established ability to regulate abiotic and biotic stress responses in plants (Baillio et al. 2020; Rushton et al. 2010; Satapathy et al. 2018; Wang et al. 2020). Our collective findings supported the selection of *SbWRKY86* for further functional characterization in plant defense responses relevant to SCA attack.

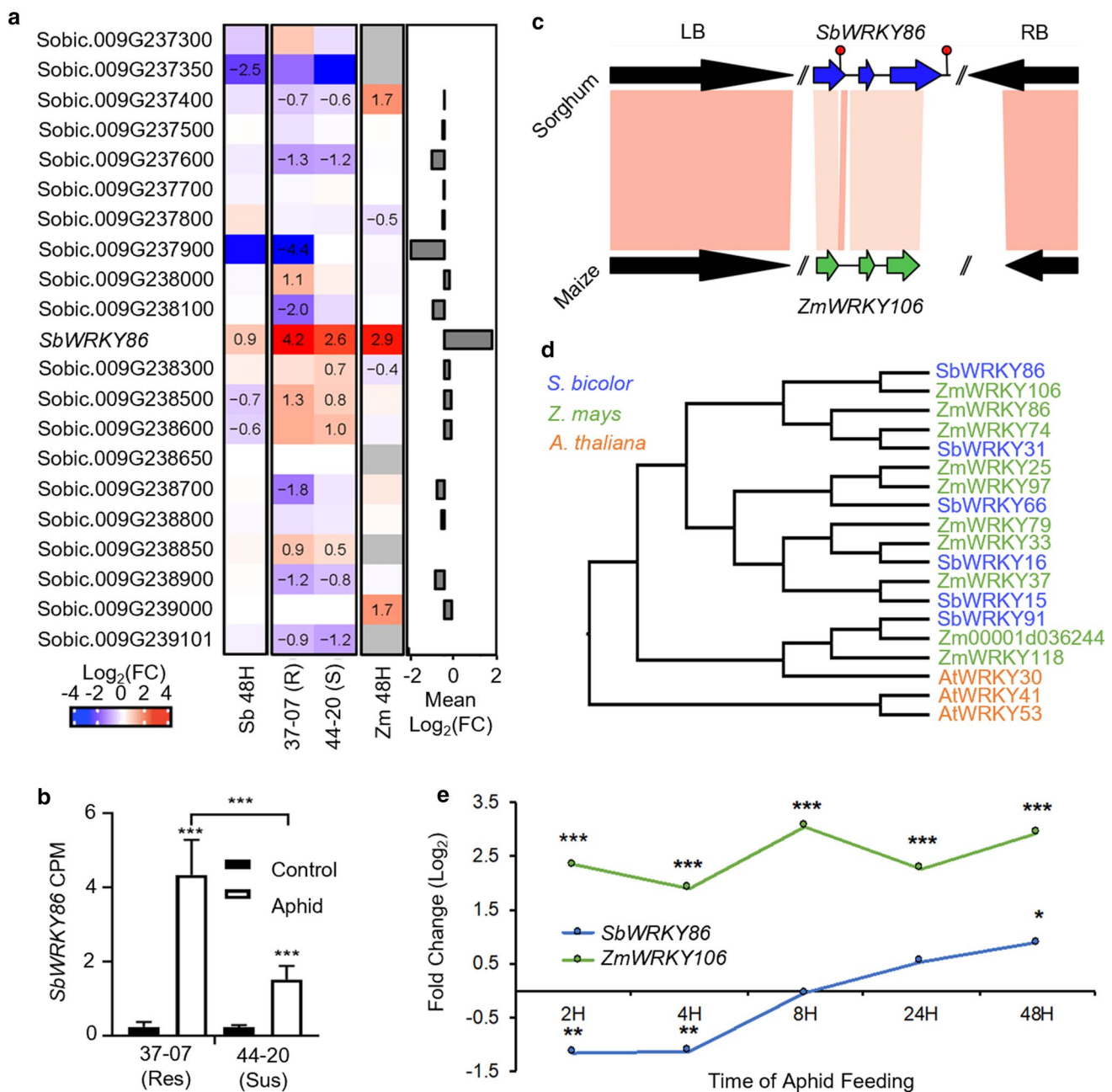
Analyses of *SbWRKY86* expression and SCA proliferation in resistant and susceptible sorghum lines

Based upon field ratings for SCA damage, we selected defined sorghum germplasm for replicated analyses in follow-up greenhouse studies. Specifically, two SCA-susceptible (PI534135, PI597950) and two SCA-resistant (PI656011, PI656043) lines were used to further consider field resistance observations and patterns of *SbWRKY86* gene expression. Three days post-infestation (dpi), SCA populations began increasing in all lines (Fig. 3a), but population levels increased more slowly on resistant lines (PI656043 and PI656011) than on susceptible lines (PI534135, PI597950). Among the two SCA-susceptible lines, PI534135 had higher

rates of SCA proliferation than PI597950 (Fig. 3a). Earlier reports have shown that growth of *M. sacchari* populations on sorghum depends on both the plant growth stage and variety used (Armstrong et al. 2015; Neupane et al. 2020). In these studies, SCA-resistant sorghum genotypes were observed to have lower aphid densities compared to the susceptible varieties under both greenhouse and field conditions (Kiani and Szczepaniec 2018; Szczepaniec 2018), and our results corroborate these findings. To examine whether *SbWRKY86* expression differed between the SCA-resistant and susceptible lines, relative expression was quantified using qRT-PCR before and after SCA infestation as compared to the housekeeping gene glyceraldehyde 3-phosphate dehydrogenase (*GAPDH*). In un-infested control plants, SCA-resistant genotypes displayed approximately twofold higher basal levels of *SbWRKY86* expression than SCA-susceptible genotypes (Fig. 3b). After 48 h of SCA herbivory, expression levels of *SbWRKY86* increased significantly in the resistant line, PI656011 as compared to the susceptible lines, PI534135 and PI597950 (Fig. 3c). On average, transcript levels of *SbWRKY86* increased in all lines following SCA herbivory but both average basal and induced *SbWRKY86* transcript expression levels were higher in SCA-resistant than -susceptible lines. These results are consistent with a hypothesis that elevated *SbWRKY86* expression present in resistant lines correlate with reduced aphid SCA proliferation and plant injury levels.

Heterologous expression of *SbWRKY86* increases aphid resistance in *Nicotiana benthamiana* and *Arabidopsis thaliana*

To further examine a potential role for *SbWRKY86* expression in mediating aphid resistance, we generated a CaMV 35S promoter-driven yellow fluorescent protein (YFP)-tagged fusion construct. The 35S::*SbWRKY86*-YFP construct was expressed transiently in *Nicotiana benthamiana* through *Agrobacterium*-mediated transformation to assess effects on aphid resistance. Given the comparatively narrow SCA host range consisting mostly of grass species, we examined resistance using a tobacco-adapted strain of the green peach aphid (*Myzus persicae*) (Ramsey et al. 2014). In *N. benthamiana* leaves expressing *SbWRKY86*-YFP, a significant reduction in *M. persicae* proliferation ($P < 0.0001$) was observed 7 days post-infestation (dpi) as compared to leaves expressing an empty vector construct (Fig. 4a). This reduced *M. persicae* proliferation co-occurred temporally with *SbWRKY86*-YFP protein expression as verified by western blotting (Fig. 4b) in the same cohort of plants used for bioassays. To assess whether heterologous expression of *SbWRKY86* could confer aphid resistance in a second system, we generated



stable transgenic lines of *Arabidopsis thaliana* (Col-0) expressing *p35S::SbWRKY86-YFP*. Homozygous plants for three independent insertion events, lines 10, 13 and 14, were generated and *SbWRKY86-YFP* protein expression confirmed by western blotting (Fig. 4c). In an assessment of *M. persicae* proliferation, significant reduction in aphid densities were observed in the transgenic lines 10 ($P < 0.0002$), 13 and 14 ($P < 0.0001$) as compared to the wild-type (Wt) control (Fig. 4d). Our results further support the hypothesis that the elevated expression of *SbWRKY86* can reduce aphid population increases.

Network analysis of gene expression changes in *A. thaliana* lines expressing *SbWRKY86-YFP*

To identify potential mechanisms underlying *SbWRKY86*-conferred aphid resistance, we used RNA-Seq to compare global basal gene expression in two independent *A. thaliana* lines heterologously expressing *SbWRKY86-YFP* (lines 13 and 14) as compared to Col-0 wild type (Wt) controls (Supplementary Table 9). Statistical analyses revealed 87 differentially expressed genes (DEG) that were common among the two *35S::SbWRKY86-YFP* lines, with 64 upregulated and 23 downregulated (Supplementary

Fig. 2 Comparative genomic and transcriptomic analyses of genes contained in the chromosome 9 SCA-associated locus. **a** Heatmap of RNA-Seq-derived gene expression data with values expressed as Log₂(Fold change, FC) as compared to controls. The left-most column labeled as “Sb 48H” represents data obtained in this study for gene expression in sorghum line BTx623, 48 h post-infestation with SCA as compared to un-infested control plants. The center two columns represent sorghum expression data obtained 24 h post-SCA infestation in an SCA-resistant line, 37-07 (R) versus an SCA-susceptible line, 44-20 (S) (Kiani and Szczepaniec, 2018). The right-most column labeled as “Zm 48H” represents expression data from maize for syntenic orthologue genes contained in the corresponding region of the maize genome (as defined in Supplementary Table 6); expression values are fold change 48 h post-infestation with corn leaf aphid as compared to un-infested leaves (Tzin et al. 2015). All heatmap cells containing a numerical value represent those with significant differential expression (adjusted $P < 0.05$). The adjacent horizontal bar graph represents the mean Log₂(Fold Change, FC) for each gene or syntenic orthologue gene across all experiments. **b** RNA-seq expression of SbWRKY86, expressed as counts per million reads mapped (CPM), 24 h post-SCA infestation versus un-infested controls in the SCA-resistant line, 37-07 (Res) and SCA-susceptible line, 44-20 (Sus) (Kiani and Szczepaniec, 2018). **c** Comparison of sorghum *SbWRKY86* with the corresponding maize syntenic orthologue *ZmWRKY106*, including the neighboring genes upstream (LB—left border) and downstream (RB—right border) for reference. The location of the SNPs found to be most significantly associated with SCA resistance are marked as red dots and shown relative to the *SbWRKY86* gene exons. **d** Phylogenetic tree of the clade containing *SbWRKY86* obtained from the complete tree of predicted WRKY proteins based on PlantTFDB annotation of the sorghum (*S. bicolor*), maize (*Z. mays*) and Arabidopsis (*A. thaliana*) proteomes (Complete protein-derived phylogeny and all associated genes are found in Supplementary Fig. 2 and Supplementary Table 6). **e** Comparison of time-course RNA-Seq data from sorghum line BTx623 and maize line B73 (Tzin et al. 2015) showing the Log₂(Fold Change, FC) in expression for *SbWRKY86* and *ZmWRKY106* genes post-aphid infestation versus un-infested control plants. For all experiments, the adjusted P values were calculated by the ExactTest edgeR function (*, **, *** for $P < 0.05$, < 0.01 , < 0.001)

Table 10). While it is not possible to determine from this RNA-Seq data whether the *SbWRKY86*-responsive genes are direct *SbWRKY86* targets or indirectly regulated through downstream transcription factor networks, the promoters for upregulated genes were examined for enrichment in possible *SbWRKY86*-binding response elements. A W-box sequence was predicted to be bound by *SbWRKY86* protein based on the PlantTFDB-annotated DNA-binding motif of AtWRKY30 (AT5G24110), the closest Arabidopsis ortholog to *SbWRKY86* (Jin et al. 2017). The reverse-complement of this motif closely aligns with the canonical WRKY transcription factor family DNA W-Box binding motif [(T)TGAC(C/T)] (Ulker and Somssich 2004). Promoters for *SbWRKY86*-responsive genes were significantly enriched in this motif when compared to all Arabidopsis gene promoters (Supplementary Fig. 4a, Supplementary Table 11). Of the 64 genes upregulated in *SbWRKY86*-YFP-expressing lines, promoters for 53 contained between 1 and 8 W-box motifs with an

average of 2.85 per promoter (Supplementary Fig. 4b–d, Supplementary Table 11).

Using the *SbWRKY86*-responsive DEG, a small interaction network of approximately 40 co-regulated genes was identified using STRING (Fig. 5; Supplementary Table 12) (Szkłarczyk et al. 2018). This network contained genes involved in plant stress responses caused by changes in regulation of carbon/nitrogen (C/N) balance, immunity, ethylene signaling and the priming of callose production (Fig. 5; Supplementary Tables 10, 12). Cytoscape visualization of this network highlights that genes related to C/N balance responses were significantly represented. Specific genes of interest encode proteins including PHOSPHATE-INDUCED 1 (PHI-1)/EXORDIUM-LIKE1 (EXL1) (Schröder et al. 2011; Lia et al. 2018), QUA-QUINE STARCH (QQS) (Schröder et al. 2011), ARABIDOPSIS TOXICOS EN LEVADURA 31(ATL31)/CARBON/NITROGEN INSENSITIVE 1 (CNI1) (Sato et al. 2009; Sanmartín et al. 2020; Maekawa et al. 2014), and Calcium-dependent protein kinase 32 (CPK32) (Liu et al. 2017). The network also included several proteins regulating plant immune responses, including soluble N-ethylmaleimide-sensitive factor attachment protein receptor (SNARE), termed SYNTAXIN OF PLANTS122 (SYP122) (Voigt 2014; Maekawa et al. 2014), DICARBOXYLATE CARRIER 2/PLANT UNCOUPLING MITOCHONDRIAL PROTEIN 4 (DIC2/UCP4) (Czobor et al. 2019), At3G10930 (IDA-LIKE7) as a negative modulator of ROS signaling (Vie et al. 2017), MAPKKK and Ethylene Response Factors (ERFs). Defense signaling could be generally elevated in part through increases in the Leucine Rich Repeat (LRR) receptor-like protein (RLP) co-receptor termed SUPPRESSOR OF BIR1-1 (SOBIR1) (Fig. 5). Interestingly, two proteins in this network, CNI1 and SYP122 were specifically demonstrated to regulate callose deposition and cell wall appositions to affect resistance against fungal pathogens (Sanmartín et al. 2020; Maekawa et al. 2014). Collectively, our transcriptomic analyses of genes regulated by heterologous expression of *SbWRKY86* in Arabidopsis is consistent with a narrow network of processes surrounding sugar metabolism, callose production and defense that are predictable mechanisms impacting aphid resistance in sorghum.

Increased expression of *SbWRKY86* is associated with enhanced callose deposition

As phloem sap-feeding insects, aphids deploy salivary secretions to successfully colonize the plant tissues (Will and Vilcinskis 2015; Miles 1999). Plants can sense salivary components as danger signals and use an array of defense responses to prevent aphid colonization. Callose deposition is one defense utilized by plants to block sieve elements and prevent infestation by phloem-feeding insects (Will

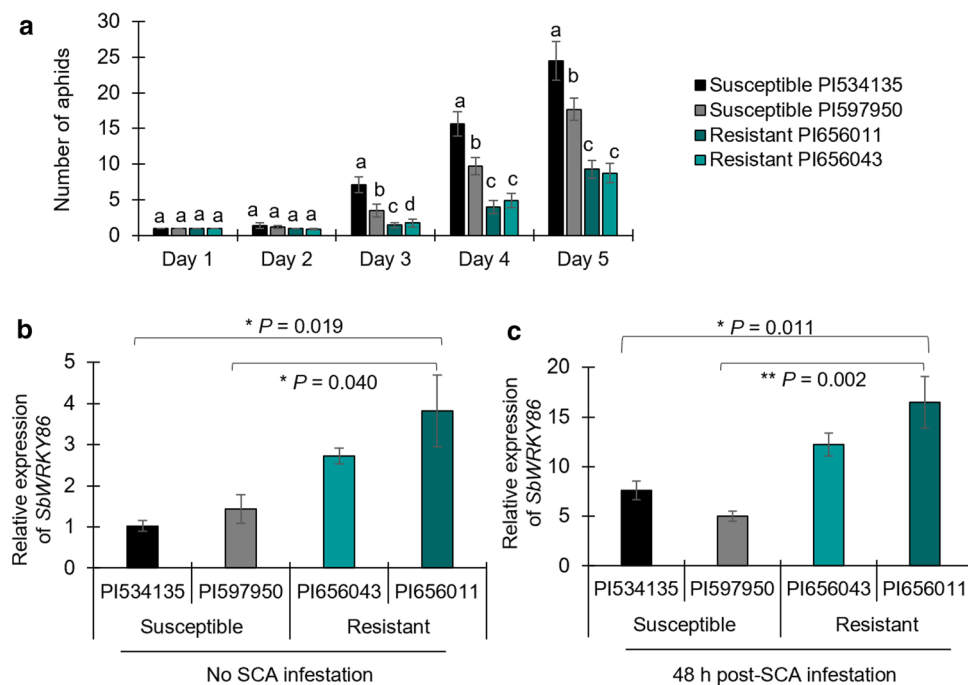


Fig. 3 Relative SCA proliferation and *SbWRKY86* expression in SCA-resistant and -susceptible sorghum lines. **a** Aphids were enumerated daily after infestation of 2-week-old plants with a single wingless adult SCA ($n=8-11$; $\pm$ SEM). Different letters (**a–d**) represent significant differences (one-way ANOVA followed by Tukey's test corrections for multiple comparisons, $P<0.05$) between the sorghum lines within each day. **b** Average ($n=4$; $\pm$ SEM) relative

expression levels of *SbWRKY86* gene expression analyzed by qRT-PCR in SCA-susceptible and -resistant lines of sorghum before and **c** 48 h after aphid infestation. Expression of the housekeeping gene *GAPDH* was used to normalize all data. For both *SbWRKY86* expression experiments (**a** and **b**), P values indicate significance within each graph as determined by a Student's t test (two-tailed distribution, unpaired)

and van Bel 2006). Earlier reports have demonstrated the role of callose deposition in resistance to phloem-feeding insects in numerous plants including rice (Hao et al. 2008), Arabidopsis (Mondal et al. 2018; Kempema et al. 2007; Casteel et al. 2014) and maize (Varsani et al. 2019). Our transcriptomic analyses of *SbWRKY86*-sensitive genes in Arabidopsis provided insights into the potential role of *SbWRKY86* as a regulator of key processes likely to impact aphid feeding, including callose deposition (Fig. 5, Supplementary Tables 10 and 11). To empirically test our hypothesis that heterologous *SbWRKY86* expression alters callose deposition, leaves were treated with flg22, a potent elicitor of immune responses in Arabidopsis, including callose deposition (Luna et al. 2011). In comparison to Wt Col-0 leaves, leaves from 35S::*SbWRKY86*-YFP lines Sb-13 and Sb-14 had increased callose production when visualized via staining and microscopy (Fig. 6a). Quantification of callose deposits confirmed that both 35S::*SbWRKY86*-YFP lines had significantly greater punctate callose deposits ($P<0.05$) in comparison to Col-0 Wt (Fig. 6b). To assess relative callose production in an SCA-resistant versus SCA-susceptible sorghum line, callose deposits were measured in leaves of the SCA-susceptible line PI 597950 versus the SCA-resistant line PI 656043 3 days post-infestation with 10 wingless adult

SCA. Callose deposits in the resistant line were significantly increased after aphid infestation ($P=0.015$), with approximately double that of the susceptible line (Supplementary Fig. 5). Our results are consistent with the hypothesis that *SbWRKY86* can promote enhanced callose deposition, a general plant immune response with established relevance to aphid feeding.

Discussion

Within the past decade, SCA has emerged as a dominant and damaging pest on sorghum in the United States (Armstrong et al. 2015). To understand genetic variation and mechanisms in sorghum germplasm potentially useful for breeding enhanced SCA resistance, multiple research efforts have identified a genetically variable cluster of NBS-LRR resistance genes on chromosome 6 at a locus termed *RMES1* (Zhang et al. 2021; Wang et al. 2013, 2021; Muleta et al. 2021). Given the damaging negative impacts of SCA on sorghum production and the need to broadly understand genes contributing to SCA resistance, we undertook a large GWAS to link SCA damage levels in the field with genetic loci and to consider additional candidate genes for characterization.

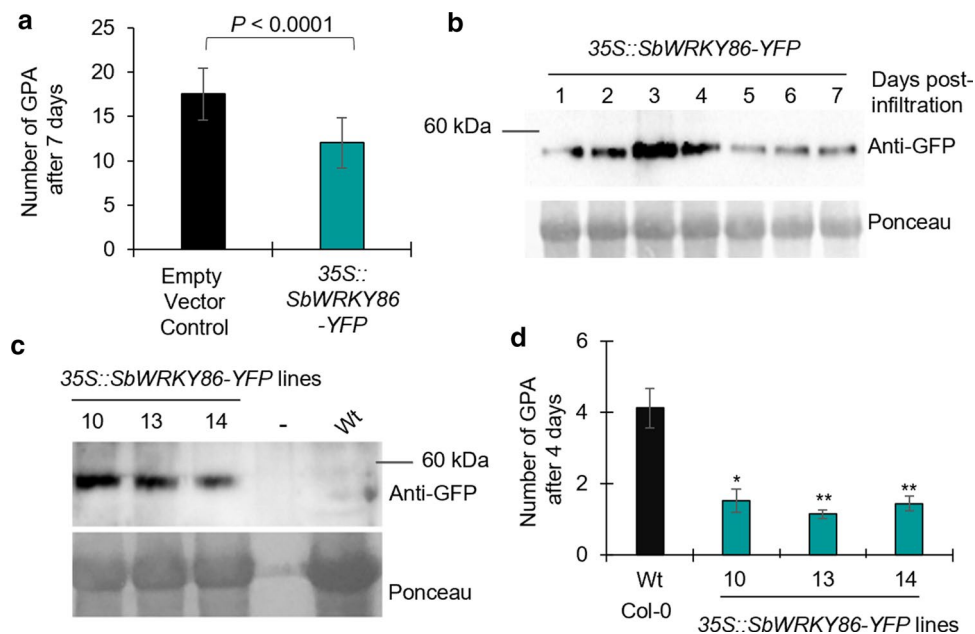


Fig. 4 Heterologous expression of *SbWRKY86* in *Nicotiana benthamiana* and *Arabidopsis thaliana* confers increased resistance to green peach aphid (GPA, *Myzus persicae*). **a** Average ($n=15$; $\pm$ SEM) GPA 7 days after *N. benthamiana* leaves expressing *p35S::SbWRKY86-YFP* were infested with a single wingless GPA adult confined by a clip-cage. *N. benthamiana* leaves expressing an empty vector construct were used as controls. **b** Expression of *SbWRKY86-YFP* protein in agroinfiltrated *N. benthamiana* leaves was confirmed for the time-course spanning the GPA bioassay, with detection using an anti-GFP antibody and Ponceau staining used to confirm equal protein loading. **c** Expression of *SbWRKY86-YFP* protein in independent lines (10,

13, 14) of *Arabidopsis thaliana* expressing *35S::SbWRKY86-YFP* detected using anti-GFP antibody and with protein loading assessed via Ponceau staining. **d** Aphid enumeration 4 days post-infestation with a single alate GPA adult for each of the three independent *A. thaliana* lines expressing *SbWRKY86-YFP* as compared to wild type (Wt) Col-0 control plants. For experiments shown in **a** and **d**, error bars represent SEM, $n=15$ and $n=25$, respectively, with a single asterisk (*) indicating a significant difference of $P<0.0002$ and asterisks (**) indicating $P<0.0001$ as determined by a Student *t*-tests (two-tailed distribution, unpaired) within each graph

In the current effort, we identified a chromosome 9 locus containing the WRKY transcription factor (TF) gene *SbWRKY86*. Given the diverse WRKY TFs known to contribute to aphid resistance, attributes of neighboring genes at the locus, and previous studies on DEGs in SCA-resistant sorghum germplasm, we characterized processes regulated by *SbWRKY86* to examine the potential for functional roles in aphid resistance. As a transcriptional regulator, the stable heterologous expression of *SbWRKY86* proteins in *Arabidopsis* reduced aphid (*M. persicae*) population increases, activated a narrow network of genes connected with C/N metabolism and immunity, and enhanced inducible callose production. Together this establishes a place for *SbWRKY86* on the growing list of WRKY TFs in diverse plant models that can influence aphid population increases (Atamian et al. 2012; Foyer et al. 2015; Kloth et al. 2016; Li et al. 2015; Shah et al. 2020; Van Eck et al. 2010; Zhang et al. 2020; Zhao et al. 2019).

Genome-wide association studies (GWAS) provide an unbiased discovery approach to statistically link genetically variable traits to SNPs and ultimately underlying gene candidates. For crop plant research, GWAS is now a mature

research tool with more than 1000 studies completed and dozens of new efforts on rice, maize and sorghum annually (Fang and Luo 2019; Liu and Yan 2019). As an important drought-resistant crop, sorghum resources have been created that now span over 10,000 germplasm lines genotyped for unified and large-scale SNP datasets (Hu et al. 2019). Traits present in sorghum seed have been used in GWAS studies to reveal gene candidates regulating levels of seed protein, fat, starch, and polyphenols along with many other traits (Boyles et al. 2016; Brenton et al. 2016; Fernandez et al. 2015; Morris et al. 2013; Rhodes et al. 2014; Sukumaran et al. 2012). To leverage these sorghum genetic resources and create a diverse, yet manageable field panel, we combined the previously genotyped SAP (Morris et al. 2013; Casa et al. 2008) and BAP (Brenton et al. 2016), resulting in 696 lines for SCA damage assessment. Across different statistical models, including the fixed and random model circulating probability unification (FarmCPU) model (Kaler et al. 2020), the SNP marker most significantly associated with SCA resistance was in the WRKY TF gene, *SbWRKY86* (Fig. 1). Importantly *SbWRKY86* was previously implicated as a DEG with elevated transcript levels in the industry SCA-resistant line

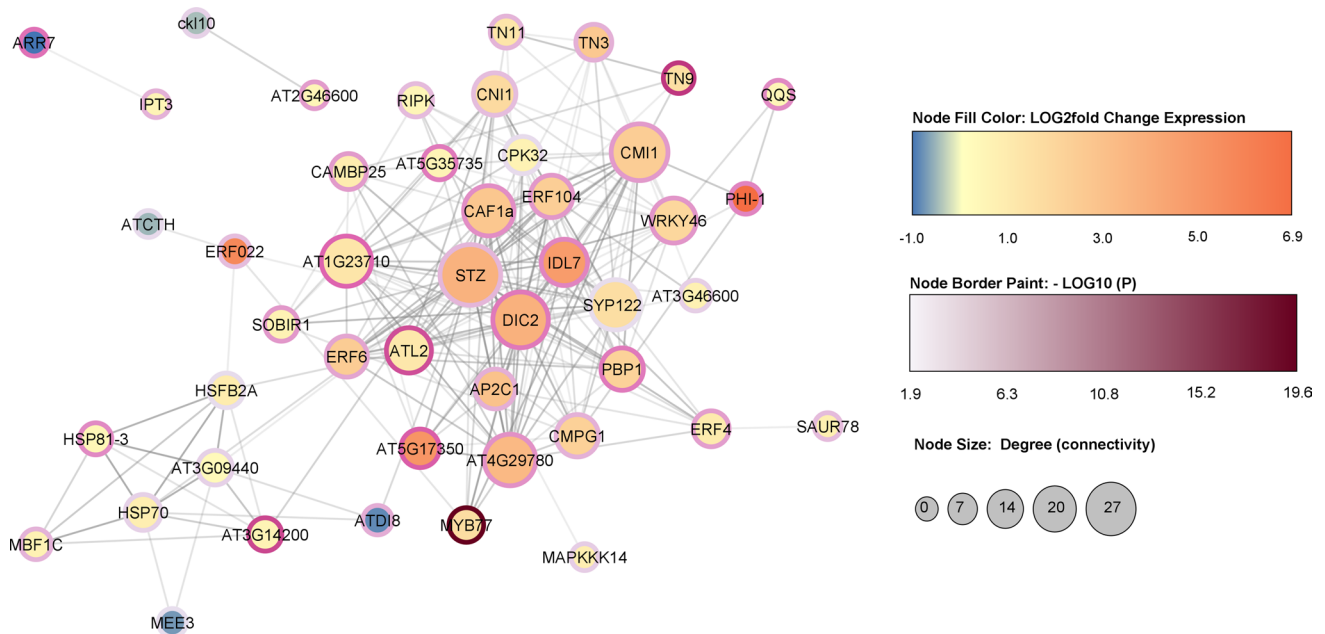


Fig. 5 Network analysis of transcriptional changes associated with heterologous expression of *SbWRKY86-YFP* in *A. thaliana*. The protein–protein interaction network was constructed using STRING analysis of predicted proteins encoded by genes found to be differentially expressed through RNA-Seq transcriptional profiling of global basal expression in *A. thaliana* 35S::*SbWRKY86-YFP* lines Sb-13 and Sb-14 as compared to wild-type Col-0 plants. Depth of the node color reflects the magnitude of log2 fold-change in expression between

Col-0 and *SbWRKY86*-expressing lines; color gradient of the node border represents the log10 (*P* value) for statistical significance in differential expression as determined using Fisher's method. Degree of connectivity and the strength of correlations are represented by the node size and edge width respectively, with a wider edge corresponding to a higher correlation. A full listing of Arabidopsis gene IDs and annotations associated with the network are detailed in Supplementary Table 9 and 10

DKS 37-07 (Kiani and Szczepaniec 2018) (Fig. 2). Transcript levels of a maize WRKY syntenic to *SbWRKY86*, *ZmWRKY106*, have also been observed to be highly expressed following aphid (*Rhopalosiphum maidis*) attack (Fig. 2, Supplementary Tables 5 & 7) (Tzin et al. 2015). RNA-seq experiments using the genome-sequenced SCA-susceptible sorghum line BTx623 demonstrated comparatively modest increases in *SbWRKY86* following sustained SCA feeding (Fig. 2; Supplementary Table 4). Our current efforts did not provide additional support for a common role of the *RMES1* locus on sorghum chromosome 6 (Wang et al. 2013; Zhang et al. 2021) (Fig. 1). This is likely due to fact that key SNPs (S6_2995581) at the *RMES1* locus associated with SCA resistance are globally rare in sorghum germplasm (Muleta et al. 2021). Instead, we selected the chromosome 9 candidate gene *SbWRKY86* for further characterization.

Genes underlying the regulation of plant defense against insect herbivores have been closely examined in diverse experimental systems (Howe and Jander 2008). In intensively examined models such as Arabidopsis, WRKY TFs have been routinely observed as an enriched class of upregulated DEG following attack by phloem feeding insects (Foyer et al. 2015). Specific WRKY TFs have been previously demonstrated to broadly regulate plant defense responses

impacting bacteria, fungi and insects (Pandey and Somssich 2009; Jiang et al. 2017). In Arabidopsis, *AtWRKY45* overexpression (Shah et al. 2020), and insertional knockout mutations of both *Atwrky20* (Zhao et al. 2019) and *Atwrky22* (Kloth et al. 2016) all resulted in significantly improved resistance to *M. persicae*. Both *Atwrky20* and *Atwrky22* knockout plants displayed signatures of enhanced salicylic acid (SA) synthesis and of SA-signaling outputs, including elevated transcript accumulation of *PATHOGENESIS-RELATED GENE 1* (*AtPRI*) (Pieterse et al. 2012), that correlated with suppressed *M. persicae* population increases (Kloth et al. 2016; Zhao et al. 2019). Arabidopsis lines impaired in SA accumulation (salicylate hydroxylase; *nahG*) or SA perception (*NONEXPRESSER OF PR GENES 1*; *npr1*) display reduced populations of both *M. persicae* and the cabbage aphid (*Brevicoryne brassicae*) (Mewis et al. 2005). Given the subtle yet complex roles of phytohormones and TFs in tuning plant responses to aphid feeding (De Vos et al. 2005), it is currently not possible to say whether native endogenous expression differences in specific TFs could be driving either susceptibility or resistance.

The sorghum SCA resistance-associated candidate gene *SbWRKY86* is a member of a small subclade of WRKYs that also includes *AtWRKY30*, *AtWRKY41* and *AtWRKY53*

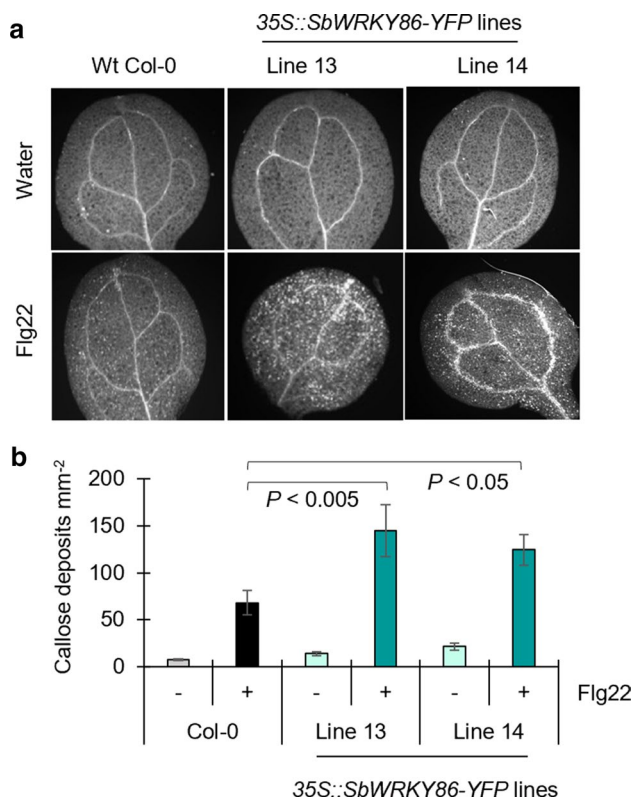


Fig. 6 Callose deposition in *A. thaliana* lines heterologously expressing *SbWRKY86-YFP*. **a** Callose deposits in leaves of wild type (Wt) Col-0 and two independent lines of *A. thaliana* expressing 35S::SbWRKY86-YFP, lines 13 and 14, before and 24 h after treatment with a 1 μ M solution of flg22 or water. **b** Quantification of callose deposits per mm² of leaf tissue with (+) and without (–) pretreatment of 1 μ M flg22 were calculated using Trainable Weka Segmentation (TWS) plugin for ImageJ. For all callose was visualized via aniline blue staining ($n=8$; $\pm$ SEM), with significant differences between flg22-treated samples determined by a pairwise Student's *t* test

(Fig. 2). Among these, transcriptional activation of *AtWRKY53* has been observed in multiple Arabidopsis-aphid RNA-seq studies (Kerchev et al. 2013; Kuśnierczyk et al. 2011, 2008). *AtWRKY53* transcripts also increase following Damage Associated Molecular Pattern (DAMP) signaling activation via Plant elicitor peptides (AtPeps) (Yamaguchi et al. 2010) and treatment with SA (Yu et al. 2001). Established functional roles for *AtWRKY53* are complex and include regulation of rapid responses to sugar (Chen et al. 2019), *Pseudomonas syringae* resistance (Murray et al. 2007; Hu et al. 2012), interactions with SA signaling (Sarkar et al. 2018; Miao and Zentgraf 2007) and leaf senescence (Zentgraf and Doll 2019). While a function for *AtWRKY53* in aphid resistance or susceptibility has not been directly examined, the most closely related ortholog in *Chrysanthemum morifolium*, termed *CmWRKY53* also accumulates at the transcript level following feeding by the aphid *Macrosiphoniella sanborni* (Zhang et al. 2020). Results from

studies of transgenic chrysanthemum plants over-expressing and under-expressing *CmWRKY53* were consistent with a negative endogenous role for *CmWRKY53* in *M. sanborni* resistance (Zhang et al. 2020). As the Arabidopsis gene most closely related to *SbWRKY86* (Fig. 2d), *AtWRKY30* exhibits strong transcriptional responses to biotic stress, promotes oxidative stress protection, heat tolerance and drought tolerance in native and heterologous systems (Scarpeci et al. 2013; El-Esawi et al. 2019). Given the pattern of higher *SbWRKY86* transcript levels associated with lower SCA populations on sorghum (Figs. 2; 3) we sought to heterologously express *SbWRKY86* in *N. benthamiana* and Arabidopsis to assess function. Both transient *SbWRKY86* expression in *N. benthamiana* and stable *SbWRKY86* expression in Arabidopsis resulted in significantly reduced population increases in *M. persicae* (Fig. 4). Our results support the hypothesis that *SbWRKY86* is the causal gene candidate underlying the initial GWAS results (Fig. 1). Given that *SbWRKY86* proteins are unlikely to be directly toxic to aphids and instead function to regulate additional gene expression networks underlying stress resistance, we conducted an RNA-seq analysis of Wt and 35S::SbWRKY86-YFP Arabidopsis plants to gain further insights into potential target genes responsive to *SbWRKY86*.

To accurately represent Arabidopsis genes transcriptionally increased by heterologous expression of *SbWRKY86*, shared responses in 2 independent lines (lines 13 and 14) were compared to Wt (Col-0) and resulted in core set of 87 DEGs (Supplementary Table 10). A STRING interaction network of 44 largely co-regulated proteins encoded by these genes (Fig. 5; Supplementary Table 12) implicates *SbWRKY86* in plant responses regulating C/N balance, immunity, signaling and priming of callose production. *PHI-1*, synonymous with *EXORDIUM-LIKE1* (*EXL1*), is directly involved in the carbon starvation response (Schröder et al. 2011). Plants with a mutant *phi-1* gene perform poorly under low light conditions and extended nights. Sorghum plants covered in SCA honeydew and adventitious sooty mold growth commonly receive reduced levels of light and thus reduced photosynthetic capacity. *PHI-1* transcripts dramatically accumulate following perception of diverse Microbial Associated Molecular Patterns (MAMPs) (Lia et al. 2018), and expression levels are known to be connected to the expression of *QQS* (Schröder et al. 2011). Overexpression of *QQS* in many plant systems increases protein levels and decreases starch levels. As part of the *SbWRKY86*-regulated STRING network, *CNII* is a RING type ubiquitin ligase that functions in the C/N stress response, acting in part during growth phase transition in Arabidopsis seedlings (Sato et al. 2009; Maekawa et al. 2014). *CNII* mRNA is cell-to-cell mobile, enabling coordination between adjacent cells. In Arabidopsis, *CNII* has been demonstrated to play a key role in the priming of callose accumulation (Sanmartín et al.

2020). As a physical means of disrupting the flow of phloem to aphid stylets, inducible callose deposition is a classical defense response used for understanding aphid resistance (Jaouannet et al. 2014). In Arabidopsis, two homologous SNAREs, SYP121 and SYP122, are established CNII-interacting proteins that regulate the formation of cell wall appositions, termed papillae, to prevent infection by biotrophic fungi (Maekawa et al. 2014). In barley, resistance to the Russian wheat aphid has been associated with enhanced papillae formation (Belefant-Miller et al. 1994), and elicited callose production and papillae responses are highly inter-related components of biotic stress protection (Voigt 2014). SNARE proteins such as SYP122 also have broader functions in vesicle fusion, and contribute to homeostasis, defense, cell expansion, and growth in plants (Zhang et al. 2008). In addition, present in the *SbWRKY86*-regulated network is CPK32, a calcium-dependent protein kinase associated with signaling in the coordination of plant C/N balance (Liu et al. 2017). Additional genes regulated by *SbWRKY86* are involved in biotic stress protection and signaling such as *ERF6* (Fig. 5) with important roles in jasmonate signaling and *Botrytis cinerea* resistance (Moffat et al. 2012), and the LRR-RLP co-receptor *SOBIR1* (Table S10) that can aid in plant perception of insect herbivory (Steinbrenner et al. 2020). To empirically test if the network of *SbWRKY86*-regulated transcripts in Arabidopsis impacts callose production, we used the defined MAMP elicitor flg22 (Mason et al. 2020). Quantification of callose deposits following flg22 elicitation supported predictions from the RNA-seq analyses (Fig. 5), demonstrating that heterologous expression of *SbWRKY86* enhances defense responses such as inducible callose production (Fig. 6). Similarly, an SCA-resistant sorghum line with higher *SbWRKY86* expression levels than an SCA-susceptible line also displayed increased callose deposition upon SCA infestation (Supplementary Fig. S5). It is increasingly accepted that callose deposition is a nearly universal mechanism that promotes phloem clogging after penetration by aphid stylets (Silva-Sanzana et al. 2019). Together, the genes regulated by *SbWRKY86* indicate several potential mechanisms by which this transcription factor could contribute to resistance against aphid attack in diverse plant systems (Supplementary Fig. 6).

In the current effort, we describe an approach to gene discovery using GWAS and rating of SCA damage levels in the field followed by an assessment of gene candidates from the locus most statistically significant associated with SCA resistance. Our collective results support the current hypothesis that enhanced expression of *SbWRKY86* can mitigate increases in sorghum SCA population levels. While heterologous *SbWRKY86* expression studies in Arabidopsis support enhanced protection against *M. persicae*, sorghum over-expression studies and assays

with SCA will be required to critically assess function in the native system. Likewise, RNA-seq or targeted qPCR analyses can be used with diverse sorghum germplasm to assess the frequency in which increased *SbWRKY86* transcript levels are associated with germplasm displaying SCA resistance. Given that plant antiherbivore defenses are highly complex (Howe and Jander 2008), it is unlikely that this mechanism will be common to all sorghum lines displaying SCA resistance. For example, important SCA tolerance mechanisms involving reduced plant damage, but higher aphid levels involve complex differences in phytohormones (Grover et al. 2020) and are unlikely to involve *SbWRKY86*. Upon further positive assessment in sorghum, SCA-resistant germplasm and molecular markers for consistently higher expression levels in *SbWRKY86* could be directly tracked in breeding programs. Successful plant defense against aphids is driven by a highly complex network of genes, proteins and metabolites (Nalam et al. 2019; Zust and Agrawal 2016). We describe *SbWRKY86* as a member of a continually growing family of WRKY TFs that can functionally fine-tune plant immune responses and alter interaction outcomes with aphids (Atamian et al. 2012; Foyer et al. 2015; Kloth et al. 2016; Li et al. 2015; Shah et al. 2020; Van Eck et al. 2010; Zhang et al. 2020; Zhao et al. 2019).

Author contribution statement AH, EAS and XN conceived the research hypotheses and designed experiments. Experiments were conducted by AH, EAS, XN, SP, EP, KD, MR, ES, AV, SW, AEC, SD and JHL, with data analyzed by AH, EAS, XN, SW, SP, KD and EP. AS and GM provided raw RNA-Seq and combined sorghum association panel SNP data, respectively. The manuscript was written by AH, EAS, SP and EP with input from all the authors.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00425-021-03814-x>.

Acknowledgements We thank Dr. Georg Jander (Boyce Thompson Institute, Ithaca, NY) for sharing his colony of tobacco-adapted *Myzus persicae*. Funding for this research was supported in part by the Department of Energy (DOE) Joint Genome Institute (JGI) Community Science Program (CSP) Project # 503420 (to A.H. and E.A.S) and the USDA-ARS NACA #58-6048-7-037, which was part of the research project entitled “Areawide Pest Management of Invasive Sugarcane Aphid in Grain Sorghum (62048-21220-018-00D).” Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture.

Data availability All data generated or analyzed during this study are included in this published article. Raw reads for RNA-Seq data generated in this study have been deposited in the NCBI Sequence Read Archive (SRA) and are available under the BioProject accession numbers PRJNA786556 and PRJNA786558.

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